Juan David Muñoz Bolaños
Rechenauerstraße 42, Innsbruck, Österreich
+43 676 9115145
jmunozbolanos@gmail.com

April 25, 2026

Spray Logics GmbH
Bleichenweg 13b
6020 Innsbruck
Austria

Unsolicited Application as Optical System Engineer / Vision Engineer
Datum: April 25, 2026

Dear Hiring Team at Spray Logics,

I have been following your innovations in multispectral sensing and automation right here in Innsbruck. As an Optical System Engineer specializing in optomechatronics, I am submitting this unsolicited application to show my interest to contribute to your sensor modules.

Currently finishing my PhD at the Medical University of Innsbruck, I specialize in the optical, mechanical, and electronic integration of optical systems. My expertise in optical spectroscopy and hyperspectral data analysis allows me to design vision systems that operate reliably under challenging environmental conditions, such as the high dust levels and fluctuating light typical in agriculture by means of adaptive data processing and machine learning. My background in hardware integration is particularly relevant: I have built automated platforms for real time control of optoelectronic hardware, including CCD cameras, FPGA Labview systems, spectrometers, single detectors, lasers, and piezo stages, ensuring high precision in the assembly of optical systems like microscopes and holographic sensors.

Furthermore, I bring experience from Leibniz IPHT Jena in developing spectroscopic systems and multispectral data processing from biological tissue samples (cancer), which are transferable to agriculture. There I built AI driven data pipelines for automated classification, a skill set that directly aligns with your efforts in vegetation and wildlife recognition. There in 2021 I developed a Python based package to process hyperspectral data since there was no clear software to process it besides the commercial ones. I think, by adhering to the V model, we can ensure innovative concepts that are translated into robust, industrial grade solutions.

Just as curiosity, the potential for laser based weed control is compelling to me, as it represents a future where light can replace chemical use. I identify with the mission of Spray Logics and would welcome the opportunity to collaborate to you in a such ecological visions, contributing to the agricultural technological landscape in Tyrol and Austria.

I appreciate the agility of an innovative firm like Spray Logics with multispectral sensing and system integration. I truly enjoy developing potential multispectral and hyperspectral light solutions for you, and I am eager to contribute my expertise to your next generation modules. As a Colombian citizen with a valid work permit for Austria, I am well settled in Tyrol and value the local life. While I am proficient in daily German, I prefer English for technical interviews to ensure precision. Being based in Innsbruck, I am available for a meeting on short notice and I hope you find the enclosed resume and CV interesting.

Sincerely,

Juan David Muñoz Bolaños
Junior Machine Vision Engineer

Juan David Muñoz Bolaños

Junior Machine Vision Engineer

Rechenauerstraße 42, Innsbruck, Austria

+43 676 9115145
jmunozbolanos@gmail.com | LinkedIn Profile

Summary

Junior Machine Vision Engineer with a lifelong specialization in multispectral and hyperspectral camera analysis. Experienced developing automated systems with machine learning and problem solving skills. Driven by innovation and the development of sustainable solutions in Tirol.

Professional/Research Experience

Jan 2022 to Present **Med. Univ. Innsbruck** *Innsbruck, Austria*
Systems Engineer | PhD Researcher

Optoelectronic & Mechanical System Design: Engineered a closed loop optomechanical (microscope based) platform for real time wavefront correction; integrated lenses, mirrors, and illumination with FPGA based control; developed 3D optomechanical supports using SolidWorks and FreeCAD, achieving precision stability.

Wavefront Control Algorithm Development: Developed custom Python wavefield simulations and statistical data analysis for optical metrology; implemented core algorithms for holographic wavefront sensing, achieving high precision characterization of wavefront measurements and error budgets.

Aug 2019 to Dec 2021 **Leibniz IPHT** *Jena, Germany*
Master's Research Assistant | Spectroscopic Systems

Hyperspectral/Multispectral System Design: Designed and built a 1064 nm dispersive spectrometer for multispectral vision; optimized fiber coupling and signal to noise ratio to enable robust detection in high scattering environments.

AI Driven Analysis and Development of PyPI Package (SpectraMap): Developed and published SpectraMap as a custom Python package on PyPI for hyperspectral data processing; architected a pipeline for automated classification, achieving 94.6% sensitivity in identification of chemical signatures.

Industrial Validation: Performed benchmarking against clinical gold standards to ensure system reliability and repeatable resolution in real time sensing.

Mar 2018 to Mar 2019 **INTECOL S.A.S.** *Medellín, Colombia*
Junior Machine Vision System Engineer

Industrial Vision & Thermal Imaging (FLIR): Engineered vision systems for bottle labeling and quality control using Halcon and FLIR thermal cameras; achieved high reliability in liquid level, cap and seal inspection.

Autonomous Robots & Production Automation: Integrated SICK light sensors and Cognex vision systems into high speed production lines, transitioning optical sensing into 24/7 industrial environments.

Technical Competencies

Machine Vision & Multispectral Sensing: Design of active multispectral systems and custom optical modules; expertise in hyperspectral camera analysis and spectral data processing; implementation of optical systems from mechanical design, to optics design and software integration.

Hardware Integration & Control Systems: Real time control of active illumination systems; deep integration of high speed sensors (CCD, FLIR, SICK, Cognex) into automated systems; development of custom hardware interfaces and drivers using Python, C++, and LabVIEW FPGA.

Industrial R&D & Software Engineering: Awareness of V Model for robust development; architect of automated data pipelines and PyPI packages (SpectraMap pypi) for ML based classification; benchmarking of optical systems against high precision standards.

Education

Jan 2022 to Jun 2026 **Medical University of Innsbruck** *Innsbruck, Austria*
Doctor of Philosophy in Optical Metrology

2019 to 2021 **Friedrich Schiller Univ. Jena** *Jena, Germany*
Master of Science in Photonics

2012 to 2018 **University of Cauca** *Popayán, Colombia*
Bachelor of Science in Physics Engineering

Publications/Technical Evidence Awards

Awards: Selected for ZEISS SMT, ASLM, and SPIE summer schools. COLFUTURO 2019 (best Colombian professionals).

Muñoz Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* (2025).

Muñoz Bolaños, J. D. et al. Dispersive 1064 nm fiber probe Raman. *Applied Sciences* (2024).

Sohmen, M., Muñoz Bolaños, J. D. et al. Optofluidic Adaptive Optics in multiphoton microscopy. *Biomedical Optics Express* (2023).

Soft Skills

Ownership & Initiative: Lead project focus from conceptual ideas to proof of application; developed holographic sensors and spectrometers.

Teamwork & Analysis: Proven collaboration across optics, hardware, and software disciplines with structured root cause analysis of system defects.

Personal Interests

• Mountain Sports (Biking, Hiking, Skiing) • Bachata Dancing • Harmonica • Learning German

Languages

English (Fluent) · German (Intermediate B2) · Spanish (Native)

References

Prof. Dr. Monika Ritsch-Marte
Medical University of Innsbruck
monika.ritsch-marte@i-med.ac.at

PD Dr. Christoph Krafft
Leibniz IPHT
christoph.krafft@leibniz-ipht.de

AGREEMENT FOR GUESTS

to carry out a practical training

with

Mr.	**Juan David Munoz Bolanos**	**Date of birth: 1994-06-04**
temporarily living at	**Emil-Wölk-Str. 7, 07747 Jena**	**(guest),**

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2020-06-02 up to 2020-12-31 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

Mr. Juan David Munoz Bolanos has to observe the security regulations for working and other regulations valid in Institute as well as instructions of the laboratory and clean room responsible persons of the Leibniz-IPHT.

Mr. Juan David Munoz Bolanos has to treat all affairs and procedures strictly confidential. Copies, duplicates and others of business and technical information of each kind are forbidden, unless these are sanctioned by the responsible Department Heads.

Publications, reports and lectures about topics which belong together with the fields of work of Leibniz-IPHT require the previous written agreement of the Executive Board.

The liability of Mr. Juan David Munoz Bolanos is limited to cases of wilful action and gross negligence.

Mr. Juan David Munoz Bolaos is insured against accidents on the Friedrich-Schiller-University Jena, Abbe School of Photonics in the context of his/her practical course during his/her length of stay at Leibniz-IPHT.

Jena, 18 May 2020

	F. Sondermann	Juan David Munoz Bolanos
Prof. J. Popp	F. Sondermann	Juan David Munoz Bolanos
Chairman	Executive Member	Guest
of Leibniz-IPHT	of Leibniz-IPHT	

Leibniz ipht
LEIBNIZ-INSTITUT für PHOTONISCHE TECHNOLOGIEN

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Mr.	**Juan David Munoz Bolanos**	**Date of birth: 1994-06-04**
temporarily living at	**Emil-Wölk-Str. 7, 07747 Jena**	**(guest),**

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2021-01-01 up to 2021-09-30 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

Mr. Juan David Munoz Bolanos has to observe the security regulations for working and other regulations valid in Institute as well as instructions of the laboratory and clean room responsible persons of the Leibniz-IPHT.

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Jena, 13 November 2020

Prof. J. Popp Chairman of Leibniz-IPHT	F. Sondermann Executive Member of Leibniz-IPHT	Juan David Munoz Bolanos Guest

INTECOL
OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos ($1,500,000) plus three hundred thousand pesos ($300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL
NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO
Administrative and Financial Director

Integración Tecnológica Industrial Intecol S.A.S.
www.intecol.com.co
Medellin Calle 42 #63c-82
PBX. (574) 316 3322 Cell: (57) 301 430 2532

UNIVERSIDAD DEL CAUCA

ON BEHALF OF THE

REPUBLIC OF COLOMBIA

AN BY AUTHORIZATION OF THE MINISTRY OF NATIONAL EDUCATION

CONSIDERING THAT

JUAN DAVID MUÑOZ BOLAÑOS

CC 1.061.772.278 OF POPAYAN

HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS,

GRANTS THE TITLE OF

PHYSICAL ENGINEER

WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM FOR THE PROFESSIONAL EXERCISE

POPAYAN, 16 MARCH 2018

REGISTERED IN THE DIPLOMA BOOK No. 082 FOLIO 437 DIPLOMA No. 396

RESOLUTION No. R-266-07 MINUTE No. 14-07

(SIGNED)

THE PRINCIPAL OF THE UNIVERSITY

THE DEAN OF THE FACULTY

THE SECRETARY GENERAL OF THE UNIVERSITY

Juan Carlos Ojeda - Garces
TRADUCTOR E INTERPRETE OFICIAL
RESOLUCIÓN No 1171
MINJUSTICIA 1997

FRIEDRICH-SCHILLER-UNIVERSITÄT JENA
PHYSIKALISCH-ASTRONOMISCHE FAKULTÄT

URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021

Friedrich - Schiller Universität Jena
2

Prof. Dr. Christian Spielmann	Prof. Dr. Silvana Botti
Dekan	*Vorsitzende des Prüfungsausschusses*

FRIEDRICH-SCHILLER-
UNIVERSITÄT
JENA

Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994
in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden

österreich ösd INNSBRUCK schweiz deutschland

Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA

Europäisches Siegel für innovative Sprachinitiativen
1998

ÖSD ist Mitglied von

ALTE

ösd
österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994
in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum
BFI Tirol Bildungs GmbH
in Innsbruck / Österreich

bestanden

Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA

ÖSD ist Mitglied von
ALTE

Europäisches Siegel für innovative Spracheninitiativen

ösd
österreich schweiz deutschland

ID-Nummer: ZB2Ö25m40434


REPUBLICA DE COLOMBIA
PASAPORTE
PASSPORT
Tipo / Type
P
Cod. pais / Country code
COL
Pasaporte Nº / Passport No.
AV410884
Apellidos / Surname
MUÑOZ BOLAÑOS
Nombres / Given names
JUAN DAVID
Nacionalidad / Nationality
COLOMBIANA
Fecha de nacimiento / Date of birth
04 JUN/JUN 1994
Núm. personal / Personal No.
CC1061772278
Sexo / Sex
M
Lugar de nacimiento / Place of birth
LA CRUZ COL
Fecha de expedición / Date of issue
11 DIC/DEC 2018
Autoridad / Authority
G. ANTIOQUIA
Fecha de vencimiento / Date of expiry
10 DIC/DEC 2028
Firma del titular / Holder's signature
ORD AV410884 10 DIC/DEC 2028
P<COLMUNOZ<BOLANOS<<JUAN<DAVID<<<<<<<<<<<<<<
AV410884<0COL9406041M2812106CC1061772278<<56

Stadtmagistrat Innsbruck
MA II - Bezirks- und Gemeindeverwaltung
Standesamt und Personenstandsangelegenheiten
Aufenthaltsangelegenheiten
post.aufenthaltsangelegenheiten@innsbruck.gv.at

Zahl: Maglbk/58566/AUFG-NAG/3/1 – 00053337-04

Antragsprotokoll

Herr: **Muñoz Bolaños Juan David**
wohnhaft in: **6020 Pradl, Reichenauer Straße 42/Top 5**
geboren am: **04.06.1994**
Reisedokument-Nr.: **AV410884**
Nationalität: **Kolumbien**
hat am: **13.03.2026**
beim Stadtmagistrat Innsbruck, Referat Aufenthaltsangelegenheiten, einen Zweckänderungsantrag auf Rot-Weiß-Rot - Karte plus eingebracht.

Ein rechtzeitig eingebrachter Verlängerungsauftrag räumt während der gesamten Dauer des Verlängerungsverfahrens dieselbe Rechtsposition ein, die nach Erhalt des letzten Aufenthaltstitels inngehabt wurde. Dies bedeutet auch, dass bei befristet erteilten Aufenthaltstiteln mit unbeschränkten Zugang zum Arbeitsmarkt („Rot-Weiß-Rot Karte plus" und „Familienangehöriger") weiterhin während der gesamten Dauer des Verlängerungsverfahrens ein Zugang zum Arbeitsmarkt besteht. Bei darüber hinaus gehenden Fragen einer allfälligen weiterhin bestehenden Beschäftigung wenden Sie sich bitte an das Arbeitsmarktservice (AMS).

Nachzureichende Unterlagen: ***

Die Erledigung des vollständig eingebrachten Antrages wird voraussichtlich innerhalb von Wochen erfolgen.

Persönliche Abholung (erforderlich !) jedenfalls erst ab: 17.05.2026, Zimmer: .Nummer ziehen

Zur Abholung (Abholzeiten: Montag bis Freitag, 08.00 bis 11.30 Uhr) sind mitzubringen:

- **der Reisepass**
- **Aufenthaltstitel in Kartenform**

Für den Bürgermeister:

(Huter)

ausstellende Behörde:

Stadtmagistrat Innsbruck
Abteilung II
Aufenthaltsangelegenheiten

übernommen am: 13.03.2026

Unterschrift:

Bei überprüfenden Amtshandlungen drei Monate nach Ausstellung dieser Bestätigung wird um Rückfrage bei der zuständigen Aufenthaltsbehörde ersucht. (Tel.: 0512/5360-1030/1032/1034/1036)